PHENIKAA UNIVERSITY

FACULTY OF ELECTRICAL AND ELECTRONIC ENGINEERING

ADVANCED DEEP LEARNING COURSE

REAL-TIME FACIAL EMOTION RECOGNITION FOR MEDICAL AND ARTISTIC APPLICATIONS

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ADVANCED DEEP LEARNING COURSE REPORT

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Abstract

Recognizing facial expressions of emotion helps clinicians follow the emotional state of patients who cannot easily say how they feel, and helps film and theatre professionals judge acting and find the emotional moments of long recordings. Doing this by eye is slow and subjective, public models are trained mostly on Western faces, and cloud services are a privacy problem for patient video. This project builds a real-time facial emotion recognition system for Vietnamese faces that runs entirely on a consumer laptop. Faces are found with YuNet, aligned to a landmark template measured on the CK+ dataset, and classified into seven classes (angry, disgust, fear, happy, neutral, sad, surprise) by a convolutional network that reads a 48×48 grayscale crop. Three networks were compared over three seeds each (EmotionCNN trained from scratch, ResNet-18 and MobileNetV2). On the 53,088-image course dataset ResNet-18 reached 68.3 ± 1.0% test accuracy, and in subject-independent 10-fold cross-validation on CK+ it reached 94.3%. A new dataset of 16,550 faces of 352 people was extracted from Vietnamese TV dramas and split by person. Its labels are still suggestions of public models, so results on it are reported as agreement (83.6%), not accuracy. A ResNet-18 trained only on these TV faces reached 53.4% on human-labeled RAF-DB test faces; adding the course dataset's human-labeled faces raised this to 79.4 ± 0.9% (macro-F1 0.713) and surprise recall from 11% to 81%. The final model drives a browser application that opens any video or image, seeks and jumps to any time, keeps the user's time marks and scans a whole film for its strongest emotional moments, at 13–17 analyzed frames per second on 1080p video. Human review of the TV labels and pilot studies with clinicians and artists are the next steps.

Keywords: facial expression recognition, transfer learning, ResNet-18, Vietnamese faces, affective computing, healthcare, film analysis, real-time application.

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1. PROBLEM SITUATION AND PROBLEM DEFINITION

1.1. Problem situation

People show much of what they feel on their faces, and many professions depend on reading those signals well. This project starts from two of them: **medicine** and **art**. In both, emotions are read today mostly by eye, during a short meeting or while watching a recording, and the observations are rarely written down in a form that can be measured or compared. Affective computing, the study of systems that recognize and respond to human emotion [1], offers tools to make this work faster and more consistent.

Medicine. Psychiatrists, psychologists and nurses observe a patient's affect (flat, sad, anxious, irritable) as part of every mental-state examination. For patients who cannot report how they feel, such as young children, people after a stroke, people with dementia or patients in intensive care, the face is one of the few available signals of pain and distress [2]. Facial behavior has also been studied as a marker of depression [3], and children with autism have practiced recognizing expressions with assistive tools that classify faces in real time [4]. Therapy and telemedicine sessions are increasingly recorded, but nobody has the time to watch hours of video and note when a patient looked sad, frightened or relieved.

Art. Film, television and theatre are built on emotion. Directors and acting teachers judge whether an actor's expression fits the scene, editors choose takes by the strength of a reaction, and researchers study the emotional arc of stories [5] and the facial responses of audiences [6]. Finding every angry, fearful or joyful moment in a 45-minute episode by hand takes hours, and two viewers often disagree.

Automatic facial emotion recognition (FER) can do the tedious part of this work: find every face in every frame, estimate its expression, and summarize a long recording on a timeline that a professional then checks. Three problems stand in the way for a Vietnamese user:

- **Faces and data.** Public FER models are trained mostly on Western faces in web photos or on posed laboratory images, while expressions differ between cultures [7]. Vietnamese faces, TV lighting and the spontaneous expressions of Vietnamese dramas are hardly represented.

- **Privacy.** Commercial services send video to the cloud. Face images are personal data under Vietnamese law [8], and patient video must not leave the clinic.
- **Trust.** Some vendors have withdrawn emotion recognition: Microsoft began retiring the emotion attribute of its Azure Face service in 2022 [9]. A tool for professionals must show what it looked at and say when it is unsure, instead of returning a single confident label.

1.2. Problem definition

The project solves the following problem. **Given a video (a file or a webcam) or a still image, detect every face and estimate for each one the probabilities of seven expression classes**: the six basic emotions of Ekman [10] plus neutral, i.e. angry, disgust, fear, happy, neutral, sad, surprise. The system must run in real time on a consumer laptop, and present its results so that a clinician or an artist can work through a long recording: play and seek, jump to any time, set their own time marks, and find the strongest moments of each emotion.

Table 1: Requirements of the system

Aspect	Requirement
Input	Any video the browser can play (MP4, WebM), still images (JPG, PNG), a webcam; files of any length
Output	Per face: box, the 7 class probabilities, the top class, and a flag when the face is outside the conditions the model was trained for
Accuracy	Measured on faces with **human** labels; macro-F1 as the main metric so that rare emotions count as much as frequent ones
Speed	At least 10 analyzed frames per second on 1080p video with a 4 GB laptop GPU
Faces	Vietnamese adults in TV and film conditions, frontal or slightly turned
Privacy	All processing on the local computer; no upload, no account
Usability	Open, switch and seek files freely; jump by typed time; own marks; whole-film summary

Scope. The system recognizes *facial expressions*, not the feelings behind them: the same expression can come from different inner states, and the link between them depends on context and culture [11]. Its output supports the judgment of a professional and never replaces it. It is a research prototype, not a medical device.

2. STATE OF ART

2.1. Existing products/solutions

Table 2 lists the tools closest to this project. Commercial products are mature but closed, paid and usually tied to a cloud service or a license; open-source tools are free but give either action units or models trained on other populations.

Table 2: Existing products and open tools for facial emotion analysis

Tool	Type	What it does	Limits for this project
FaceReader [12]	Commercial desktop software	Basic expressions and action units; used in psychology and consumer research	Paid license; closed model; not tuned to Vietnamese faces
AFFDEX SDK [13]	Commercial SDK	Real-time multi-face expression analysis; used for media and advertising tests	License; vendor-controlled models
Azure Face emotion [9]	Cloud API	Returned emotion scores per face	Retired from 2022 for responsible-AI reasons; cloud upload
OpenFace 2.0 [14]	Open-source toolkit	Landmarks, head pose, gaze and action units	No emotion classes; rules are needed on top
HSEmotion [15]	Open-source models	EfficientNet emotion models trained on AffectNet	Western-dominated training data; used here only to suggest labels
FER+ model [16]	Open ONNX model	8-class model trained on re-labeled FER2013	64×64 web faces; weak on angry and fear in our tests

2.2. Scientific/engineering publication/whitepapers

Emotion theory. Ekman and Friesen proposed six basic emotions recognized across cultures [10] and the Facial Action Coding System that describes expressions as combinations of muscle actions [17]. Later work showed that facial movements are less universal and less specific than this view assumes [7, 11], which is why this report speaks of expressions, not emotions.

Datasets. CK+ contains posed expression sequences recorded in a laboratory [18]. FER2013 collected 35,887 web faces at 48×48 pixels, and its authors estimated human accuracy at about 65% [19]; FER+ re-labeled it with ten annotators per image [16]. RAF-DB [20] and AffectNet [21] collected faces in the wild with crowd-sourced labels. On such in-the-wild data, accuracy stays far below that on posed sets such as CK+ [22].

Methods. Deep convolutional networks dominate FER [22]. Networks pretrained on ImageNet [23] and fine-tuned on faces converge faster than networks trained from scratch; ResNet adds identity

shortcuts that make deep networks trainable [24], and MobileNetV2 uses inverted residual blocks with depthwise separable convolutions for mobile devices [25]. A FER pipeline also needs a face detector such as YuNet [26], an alignment step that maps facial landmarks onto a template with a similarity transform [27], and for dataset building a face-recognition embedding such as SFace [28] to keep the same person out of both training and test sets. Grad-CAM shows which image regions drive a decision [29].

Applications. Facial expression analysis has been used to detect pain [2], to study depression [3], to teach children with autism [4] and to measure audience responses to video [6]. These works show the value of the technology, and also that each application needs data from its own population.

Gap. No public model or dataset targets Vietnamese faces in TV and film conditions, and most published results are measured on the same kind of data the models were trained on. This project builds such a dataset, measures accuracy only on human labels, and packages the model in a local application.

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3. OBJECTIVES

3.1. General objectives

Build and evaluate a real-time facial emotion recognition system for Vietnamese faces that helps medical and artistic professionals review recordings, from data collection to a working application on a consumer laptop.

3.2. Specific objectives

1. **Data:** build a dataset of Vietnamese TV faces with a reproducible pipeline that splits by person, with at least 1,000 faces per class, and combine it with the course dataset.
2. **Models:** compare three CNNs (EmotionCNN from scratch, ResNet-18, MobileNetV2) in the same pipeline, three seeds each, with macro-F1 and bootstrap 95% confidence intervals.
3. **Transfer:** pre-train on CK+ (clean, posed labels), measure subject-independent accuracy on it by 10-fold cross-validation, and fine-tune on the target data.
4. **Honest evaluation:** report accuracy only on human-labeled faces; report results on model-suggested labels as agreement.
5. **Application:** a local application with at least 10 analyzed frames per second on 1080p video that opens any file, seeks, jumps to any time, keeps the user's marks and summarizes a whole film.
6. **Responsible use:** flag faces outside the training conditions, show the crop the model sees, keep data on the computer, and document the limits.

4. PROPOSED SOLUTION

4.1. Description of proposed solution/design

4.1.1. Block diagram

The solution has two parts. The **offline part** (Figure 2) builds the data and trains the models: a ResNet-18 pretrained on ImageNet is trained first on CK+ and then on the target faces. The **online part** (Figure 1) is the application: a page in the browser plays the file and sends frames to a local server, which runs the same face pipeline that prepared the training faces.

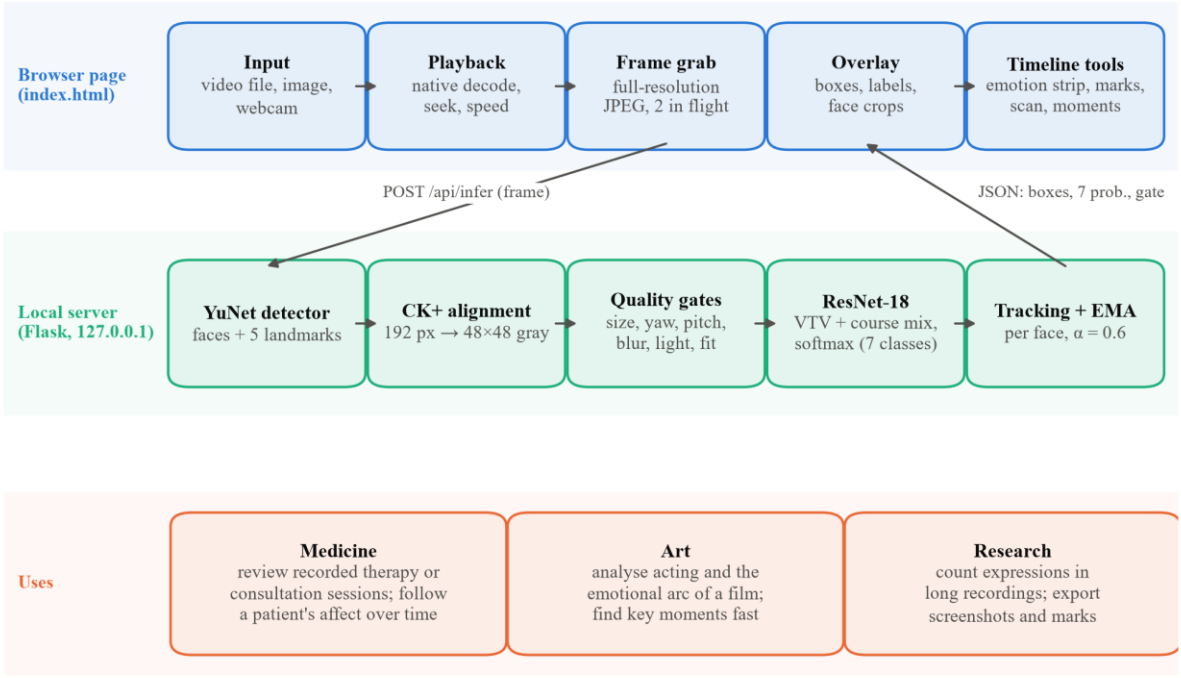


Figure 1: Block diagram of the application and its intended uses

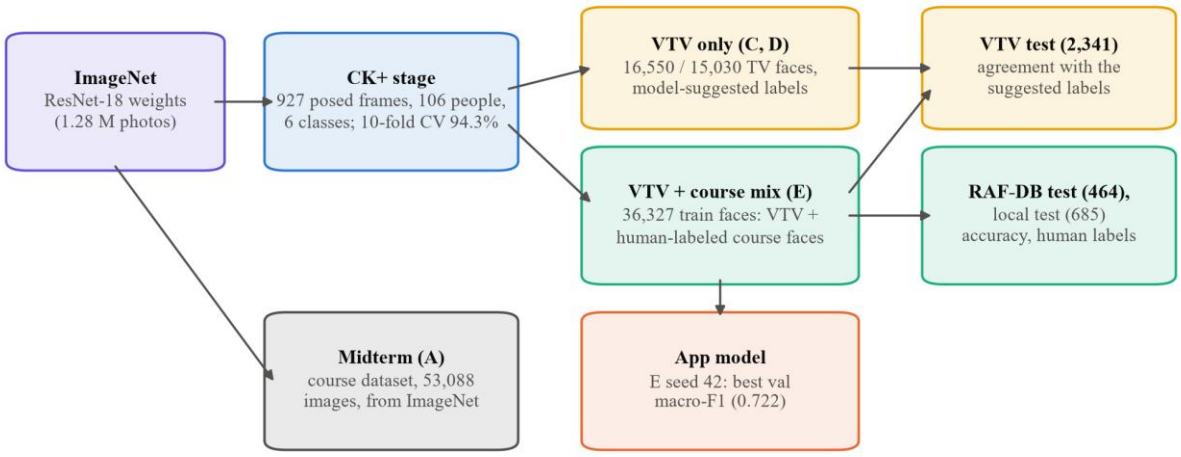


Figure 2: Training path: ImageNet → CK+ → target data (stages A–E of Chapter 8)

4.1.2. Operation description

A user opens a video or image (button, drag and drop, or the list of films on the computer). The browser decodes and plays it natively, so seeking, speed and volume work as in any video player. For every analyzed frame the following steps run:

1. **Frame grab.** The page draws the current frame on a canvas and encodes it as JPEG at full resolution. Two frames are in flight at once: while the server analyzes one, the browser encodes the next.
2. **Detection.** YuNet [26] finds the faces and five landmarks (eyes, nose tip, mouth corners) on the frame scaled to at most 960 pixels on the long side.
3. **Alignment.** The five landmarks are mapped onto the mean landmark layout of the CK+ 48×48 crops by the similarity transform of Eq. (1) [27]; the face is cut out at 192×192 pixels, converted to grayscale and resized to 48×48, exactly as the training faces were prepared.
4. **Quality gates.** Each face is measured: eye distance at least 20 pixels, yaw $|y| \leq 0.20$ (about 20°), pitch and landmark fit within the CK+ range, brightness, contrast, at most 12% outside the frame, and sharpness. A face that fails is still classified but drawn gray with the reason (turned, small, blurred, dark…), because the model never trained on such faces.

5. **Classification.** The crop goes through the model's validation transform (grayscale copied to three channels, resized to 224×224 on the GPU, ImageNet normalization) and ResNet-18; Eq. (2) turns the outputs into probabilities.
6. **Tracking and smoothing.** Faces are matched to the previous frame by box overlap and their probabilities are smoothed over time by Eq. (3), which removes flicker between frames.
7. **Display.** The server returns boxes, raw and smoothed probabilities, the gate result and the 48×48 crop as JSON. The page draws the boxes over the video, the seven bars, the face crop next to the crop the model saw, and colors the timeline by the dominant emotion of each watched part.

$$(\hat{s}, \hat{R}, \hat{t}) = \arg\min_{s,R,t} \sum_{i=1}^{5} \| s\, R\, x_i + t - y_i \|^2 \quad (1)$$

where x_i are the detected landmarks, y_i the template landmarks, s a scale, R a rotation and t a translation.

$$p_c = \frac{\exp(z_c)}{\sum_{k=1}^{7} \exp(z_k)} \quad (2)$$

$$s_t = \alpha\, p_t + (1 - \alpha)\, s_{t-1}, \quad \alpha = 0.6 \quad (3)$$

Here z are the network outputs (logits), p the class probabilities, and s the smoothed probabilities of one tracked face; $\alpha = 0.6$ gives the new frame slightly more weight than the history.

Besides the live view, the application has tools for long recordings, which are the core of the medical and artistic use:

- **Jump to any time.** A box accepts `1:23:45`, `83:20`, `300` (seconds), `50%`, and relative steps such as `+30` or `-1:00`; keys move by 5 s, 10 s, 1 min or one frame, and 0–9 jump to 0–90% of the file.
- **Own time marks.** A mark stores the current time and a label (by default the emotion of the selected face). Marks appear on the timeline and in a list where they can be renamed, moved or deleted, and are saved per file.
- **Whole-film scan.** A hidden second video element samples one frame every 1, 2, 5 or 10 seconds, so the whole timeline is colored by emotion. The six strongest moments of each

emotion (probability ≥ 0.55, at least 4 s apart) are listed with a thumbnail; a click jumps there. The scan result is saved per file.

- **Follow one face.** A click on a face pins it; the panel then shows only that person.
- **Evidence.** A screenshot with the boxes is saved to the project folder, for a case note or an editing log.

4.1.3. Proposed devices and components

Table 3: Hardware and software components

Component	Choice	Role
Laptop	ASUS TUF FX706HC: NVIDIA RTX 3050 Laptop GPU (4 GB), 16 GB RAM, Windows 11	Data pipeline, most training, the application
Cloud GPU	Google Colab, NVIDIA T4 (free tier)	CK+ cross-validation (90 runs), VTV transfer runs
Deep learning	Python 3.14, PyTorch 2.14 with CUDA 13.0 [30], torchvision	Models, training, inference
Vision	OpenCV [31]: YuNet detector, SFace embeddings	Face detection, landmarks, person grouping
Label suggestion	ONNX Runtime: FER+ and two HSEmotion models	Suggested labels for the TV faces only
Evaluation	scikit-learn [32], NumPy, pandas, Matplotlib	Metrics, splits, figures
Application	Flask (local server), HTML/JavaScript page, Chrome app window	Real-time interface
Camera (optional)	Any USB or built-in webcam	Live use

Table 4: The three compared networks (inference time per image at batch size 1)

Network	Training	Parameters	CPU (ms)	GPU (ms)
EmotionCNN (`cnn_scratch`)	from scratch, 1-channel 96×96	4.69 M	13.5	1.5
ResNet-18 [24]	ImageNet → two-stage fine-tuning, 224×224	11.18 M	18.2	2.9
MobileNetV2 [25]	ImageNet → two-stage fine-tuning, 224×224	2.23 M	13.0	5.9

EmotionCNN has four blocks of two 3×3 convolutions with batch normalization [33] and dropout [34]; the last block uses dilated convolutions to widen the receptive field without losing resolution. ResNet-18 and MobileNetV2 replace their classifier with dropout 0.5 and a 7-way linear layer. Times are from the midterm evaluation on the laptop.

4.2. Solution limitations

- **Expression is not emotion.** The model sees muscle movements in one frame; a polite smile and real joy look alike, and masked or culturally shaped expressions are missed [11].

- **Seven discrete classes.** No intensity, valence or arousal, no action units, and no pain or confusion class, which some clinical uses would need.
- **Frontal faces only.** Faces turned by more than about 20°, smaller than 20 pixels between the eyes, blurred or badly lit are flagged gray; in crowd scenes many faces are gray.
- **48×48 grayscale input.** Fine details and color cues (blushing, tears) are lost.
- **Training population.** The TV faces are actors performing emotions; the course faces are web photos. There are no patients, few children or elderly people, and no clinical validation.
- **Unchecked TV labels.** The labels of the Vietnamese TV faces are suggestions of public models; no human has checked a TV test set yet, so accuracy on Vietnamese TV faces is still unknown.
- **Frame rate.** The application analyzes 13–17 frames per second, so on 25–30 fps video it analyzes about every second frame; very short expressions can be missed.

4.3. Expected results

At the start of the final phase the following results were expected; Chapter 8 compares them with what was achieved.

- Subject-independent accuracy above 90% on CK+, where expressions are posed and clear.
- Accuracy of at least 70% on human-labeled in-the-wild faces, above the label-suggestion models.
- A Vietnamese TV dataset of more than 10,000 faces, at least 1,000 per class, with no person in two splits.
- A real-time application with at least 10 analyzed frames per second at 1080p on the laptop.
- Deliverables: dataset pipeline, trained models, application, and this report.

4.4. Design approach and implementation methodology

The work followed an iterative loop: build a baseline, measure it on human labels, find the weakest point, fix it, and measure again (Figure 2). The midterm baseline trained three networks on the course dataset; the final phase added the CK+ stage, the Vietnamese TV dataset, and the application. When

the first demo looked weak on films, the model was scored against human labels, which revealed the cause (Section 7.2.3), and the training data was fixed.

Rules applied in every experiment:

- **Fixed splits, chosen by person.** The course dataset keeps its own split; CK+ and the TV faces are split so that no person appears in two splits. Near-duplicate images across splits were removed before training.
- **Selection on validation only.** Hyperparameters, epochs and checkpoints are chosen on the validation set; each test set is used once per final model.
- **Three seeds** (42, 123, 2024) per configuration; results are reported as mean ± standard deviation, with macro-F1 (Eq. (6)) and a bootstrap 95% confidence interval [35].
- **Sanity checks** before every full run: a two-batch dry run and an overfit-one-batch test.
- **Configuration files** hold every path and hyperparameter; each run directory stores its merged config, log, best checkpoint, curves and test metrics.

Training uses cross-entropy with label smoothing [36] (Eq. (4)), AdamW [37] without weight decay on biases and batch-norm parameters, and a one-epoch linear warm-up followed by cosine decay [38] (Eq. (5)), stepped every iteration.

$$L = -\sum_{c=1}^{7} q_c \log p_c, \quad q_c = (1-\varepsilon)\,1[c = y] + \frac{\varepsilon}{7}, \quad \varepsilon = 0.1 \tag{4}$$

$$\eta_t = \eta_{min} + \frac{1}{2}(\eta_0 - \eta_{min})\left(1 + \cos\frac{\pi\, t}{T}\right) \tag{5}$$

$$F1_{macro} = \frac{1}{7}\sum_{c=1}^{7} \frac{2\,P_c\,R_c}{P_c + R_c} \tag{6}$$

In Eq. (4) y is the true class and ε spreads 10% of the target over all classes, which discourages over-confident outputs on noisy labels. In Eq. (5) η_0 is the stage's initial learning rate, $\eta_{min} = 10^{-6}$ and T the number of iterations of the stage. In Eq. (6) P_c and R_c are the precision and recall of class c.

Table 5: Training configuration of the pretrained networks

Item	Setting
Stage A (head warm-up)	Backbone frozen, classifier only; 3 epochs, learning rate 1e-3
Stage B (fine-tuning)	ResNet-18: layer3 5e-5, layer4 1e-4, classifier 5e-4; MobileNetV2: features[7:14] 5e-5, features[14:] 1e-4, classifier 5e-4; 20 epochs; earlier layers frozen
Regularization	Dropout 0.5, weight decay 5e-2 (tuned on validation: 1e-4 over-fitted, train macro-F1 0.95 vs validation 0.69), label smoothing 0.1
Batch, precision	Batch 32; mixed precision (AMP) on CUDA; gradient clipping at norm 5
Early stopping	On validation macro-F1, patience 6 epochs; best.pt = best validation macro-F1
Augmentation	Random resized crop (scale 0.85–1), horizontal flip, rotation ±10°, brightness and contrast ±0.3, random erasing (p = 0.25); applied at 48×48, then resized on the GPU
Not used	Vertical flips, rotations above 15°, CutMix (it can cut away the eyes or mouth)

5. POTENTIAL IMPACT AND CONSIDERATION

5.1. Technical viability

The whole system runs on a laptop GPU with 4 GB of memory. The final ResNet-18 needs 2.6 ms per face on the GPU and 17 ms on the CPU, and its checkpoint is 43 MB, so a clinic or studio PC without a GPU can still run it, more slowly. The application uses only a browser and a local Python server, with no internet connection. The main technical risks are the label quality of the TV data and faces outside the training conditions, both discussed in Chapter 9.

5.2. Economic feasibility

All software is open source and all compute was free (the laptop already owned and the free Colab tier), so the project cost almost nothing in cash (Chapter 10). Reviewing a recording with the application is much faster than by eye: a whole 41-minute episode is scanned at one frame every 10 seconds in about 70 seconds, after which the user jumps straight to the listed moments. For a hospital or a film school this means a free tool on existing computers instead of a paid license.

5.3. Social viability

The benefits are a tool tuned to Vietnamese faces, and time saved for clinicians, editors and teachers. The risks are misuse for surveillance, and over-trust: treating a label as a diagnosis or as proof of what someone feels. The design limits both: data never leaves the computer, every label comes with its probability and the crop the model saw, faces outside the training conditions are gray, and the documentation states that the output supports, not replaces, a professional. Use on people requires their informed consent.

5.4. Operational viability

The application starts with a double click on a launcher, opens its own window and stops when the window is closed. Files are added from a list, a dialog or by drag and drop; marks, scan results and the last position are remembered per file in the browser. The model can be switched in the window among all finished training runs. Retraining needs only the configuration files and one command per seed; the dataset pipeline is resumable and runs one process at a time so that it fits in 16 GB of memory.

5.5. Environmental feasibility

Training and data preparation used about 20 GPU hours on the laptop (Chapter 10), roughly 3 kWh of electricity, plus free Colab time. The deployed model is small, runs on existing hardware, and processes video locally instead of streaming it to a data center. No new hardware was bought.

5.6. Alternatives solution/approach

Table 6: Alternative approaches

Alternative	Advantage	Why not chosen (now)
Cloud emotion API	No training; maintained by a vendor	Uploads patient or film video; services have been withdrawn [9]; cost per call
Commercial software (e.g. FaceReader)	Validated in research, action units	License cost; closed model; not tuned to Vietnamese faces
Action units (OpenFace) + rules	Interpretable, standard FACS output	Rules for 7 classes are brittle; AU detectors also trained on other populations
Larger models (EfficientNet, ViT)	Higher accuracy on AffectNet-scale data	Slower on a 4 GB laptop; small target dataset; harder to explain
Video (temporal) models	Use motion, catch short expressions	Need labeled clips; the EMA smoothing already gives temporal stability
Multimodal (voice, physiology)	More reliable emotion estimate	Extra sensors and data; outside the course scope

5.7. Codes of ethics (IEEE)

The IEEE Code of Ethics [39] asks engineers to hold paramount the safety, health and welfare of the public, to protect privacy, to be honest and realistic in stating claims, and to treat all persons fairly. Table 7 shows how the project applies these principles. The EU AI Act [40] prohibits emotion recognition in workplaces and schools except for medical or safety reasons; this project's focus on medicine and art is consistent with that line, and uses such as employee or student monitoring are out of scope.

Table 7: IEEE ethics principles and how the project applies them

Principle	Application in this project
Safety, health, welfare	Not a medical device; output is advisory; gray flag and probability on every face; limits stated in the application guide and this report
Privacy	Local processing only; no account, no upload; screenshots only when the user asks; TV frames and datasets kept private and not redistributed
Honesty of claims	Results on model-suggested labels are called agreement, never accuracy; failed or incomplete runs are reported but kept out of the means; test sets used once
Fairness	Training data extended to Vietnamese faces; fairness across age, gender and skin tone not yet measured, listed as a risk [41]
Disclosure of tools	An AI coding assistant (Claude Code) helped write code and run experiments under the students' direction; its own visual screen of labels was tested and rejected (Section 7.1.3)
Intellectual property	Public datasets used under their research licenses; TV episodes used only for research; open-source software credited

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6. WEEKLY PROJECT DEVELOPMENT SCHEDULE

6.1. Member roles

Table 8: Roles

Person / tool	Role	Main tasks
Nguyễn Quang Trung (22011211) and Nguyễn Tiến Thành (22011103)	Students, team members	Data collection and checks, model design and training, evaluation, application, report (shared)
Ph.D Minhhuy Le	Advisor	Course supervision, review of methods and report
Claude Code (AI assistant)	Tool	Drafting code and scripts, running long jobs, figures, under the students' review
Google Colab	Tool	Free T4 GPU for the CK+ cross-validation and transfer runs

6.2. Schedule and plan

Figure 3 shows the weeks of the project, reconstructed from the run logs and file dates, and the work planned after this report. Table 9 lists the output of each week.

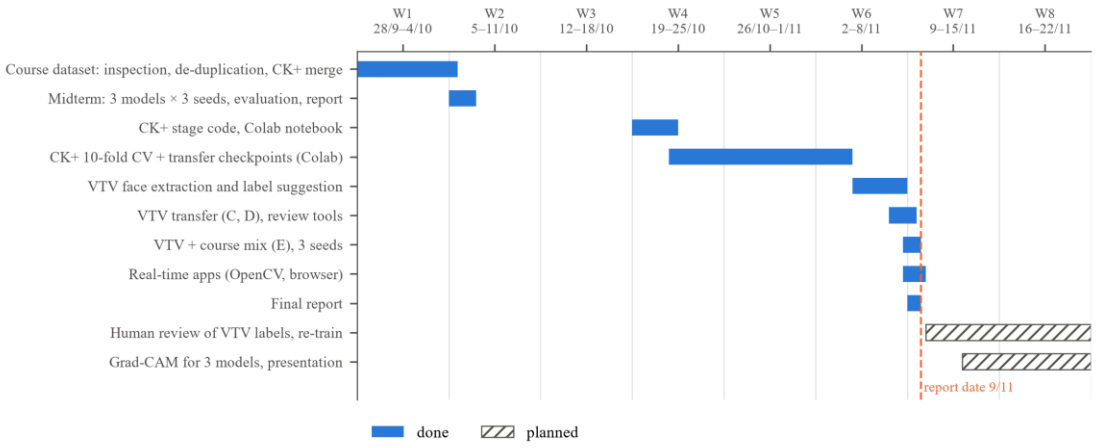


Figure 3: Project schedule (2026): done (blue) and planned (hatched)

Table 9: Weekly work and outputs

Week (2026)	Work	Output
W1, 28/9–4/10	Course dataset inspection, cross-split de-duplication, CK+ merge by person, training loop and sanity checks	Clean splits (567 near copies moved), shared code
W2, 5–11/10	Midterm: 3 networks × 3 seeds, evaluation, ensemble	Midterm report (6/10)
W4, 19–25/10	CK+ stage: folds by person, configurations, Colab notebook	CK+ pipeline (21/10)
W5, 26/10–1/11	CK+ 10-fold CV (90 runs) and 9 transfer checkpoints on Colab	CK+ results (3/11)
W6, 2–8/11	TV face extraction, label suggestion, review page, dataset build; transfers C and D; human-label check; first mix run; OpenCV demo	VTV dataset (7/11), demo (8/11)
W7, 9–15/11	Mix model with 3 seeds, browser application, this report; then human review of TV labels	Final model and application (9/11)
W8, 16–22/11	Retraining on reviewed labels, Grad-CAM for 3 networks, presentation (planned)	Presentation

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7. PROJECT PROGRESS

7.1. Topic 1: Datasets

7.1.1. Theoretical basis

Three properties of FER data decide how far results can be trusted. **Identity leakage:** if the same person is in training and test, the model can recognize the person instead of the expression, and test scores are inflated; splits must therefore be made by person, and near-duplicate images across splits removed. **Label noise:** annotators disagree on in-the-wild faces (human accuracy on FER2013 is about 65% [19]), and labels suggested by models are noisier still, so a test set with model labels measures agreement with those models, not accuracy. **Domain shift:** a model trained on posed laboratory faces or web photos loses accuracy on TV faces with other lighting, pose and population.

7.1.2. Calculation/analysis

Course dataset. 53,088 grayscale 48×48 faces in 7 balanced classes (Table 10), from four sources: a local FER2013-style collection (15,198 originals), RAF-DB (9,979), AffectNet (3,133, only angry, disgust and fear) and CK+ (922). A near-duplicate search (perceptual hash plus pixel correlation over small shifts and a flip) found validation and test images that were copies of training images; 567 files were moved out and the splits re-balanced. CK+ frames were assigned by person (train 84 people, validation 16, test 6).

Table 10: Course dataset after de-duplication

Split	Images	Per class	Content
Train	47,712	6,816	23,856 originals + one augmented copy each
Validation	4,025	575	Originals only
Test	1,351	193	Originals only

CK+. 927 frames (the last three frames of 309 labeled sequences) of 106 people in 6 classes; CK+ has no neutral class and contempt was dropped. Folds were built with StratifiedGroupKFold over people [32].

Vietnamese TV dataset (VTV). Figure 4 shows the pipeline. Six TV programs (12 video files: dramas, a comedy show and an old film) and 20,718 trailer frames were sampled at 10 frames per second. In the main run, YuNet found 414,003 faces, of which only 7,258 (1.8%) passed all gates

(Table 11): most faces were too small, turned or repeated. Every source together kept about 32.7 thousand aligned faces for review.

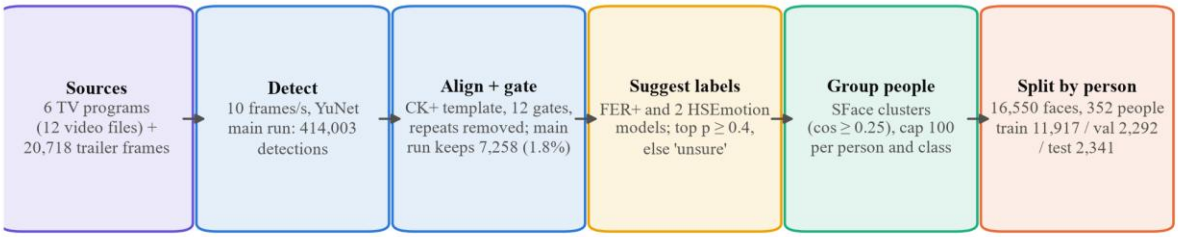


Figure 4: Construction of the Vietnamese TV face dataset

Table 11: First failed gate of the detected faces in the main extraction run

Gate	Faces	Gate	Faces
small (eye distance < 20 px)	171,031	dark or bright	4,294
detector score < 0.80	82,290	pitch	3,299
yaw (turned)	81,081	outside the frame	2,892
repeat of a kept face	32,548	landmark fit	1,293
blur	15,723	roll	1,093
contrast	10,430	recheck on the crop	771
kept	**7,258**	**total detections**	**414,003**

Labels. FER+ alone recognized only 13% of CK+ angry and 8% of CK+ fear faces, so labels are suggested by the average of FER+ and two HSEmotion models. Measured on faces prepared exactly like the TV crops, this ensemble raised accuracy from 0.659 to 0.887 on CK+ and from 0.548 to 0.670 on RAF-DB compared with FER+ alone. Faces whose top probability is below 0.4 go to an 'unsure' folder; at this threshold the ensemble sorted 84% of RAF-DB faces with 74% accuracy, and 74% of the TV faces.

People and splits. Actors appear in many episodes, so faces are grouped by person with SFace embeddings [28] and average-linkage clustering. The threshold was calibrated on the data: two faces in one frame (different people) had a median cosine similarity of 0.11, the same person in other shots 0.54–0.76. At 0.40, 6% of test faces still had five or more near-identical training faces; at 0.25, none. One comedy performer made up 14% of all faces and 86% of the test set until a cap of 100 faces per

person and class was added. The split uses StratifiedGroupKFold with 7 folds over people (test = fold 0, validation = fold 1).

7.1.3. Results

The VTV dataset has **16,550 real faces of 352 people** (Table 12); the identity leak check found 1.8% of validation and 1.3% of test faces with five or more similar training faces. Classes with fewer than 1,000 faces received 614 augmented copies of their training faces, so that every class has at least 1,000; validation and test faces are all real. Figure 5 shows test faces with their suggested labels.

Table 12: VTV dataset (suggested labels): real faces per split

Class	Train	Validation	Test
angry	788	149	156
disgust	506	85	113
fear	672	111	112
happy	2,816	563	563
neutral	4,629	926	926
sad	1,920	364	364
surprise	586	94	107
total (people)	**11,917 (247)**	**2,292 (54)**	**2,341 (51)**



Figure 5: Aligned 48×48 VTV test faces per suggested class (random sample)

A screen that hurt. To speed up the review, the AI assistant screened the four weakest classes by eye and removed 1,844 of 4,196 faces (blurred, turned, or a doubtful emotion). On 48 RAF-DB faces its eye was less reliable than the ensemble (58% vs 71%). Models trained on the screened version (D) agreed *less* with the labels on the 440 test faces both versions share (ResNet-18: 79.7% → 75.8%), with the loss in exactly the screened classes (surprise 63–76% → 26–39%). Removing faces made the rare classes too small; the unscreened version (C) remained the main TV dataset.

Mix dataset. For stage E, the course dataset's real faces (not its augmented copies) were aligned the same way as the TV faces and joined with the VTV build (Table 13). Each part keeps its own split, so the RAF-DB and local test faces stay unseen. The rare classes grow from 500–800 to about 4,200 training faces.

Table 13: Mix dataset (stage E) by source

Source	Train	Validation	Test	Labels
VTV (Vietnamese TV)	11,917 + 614 aug.	2,292	2,341	model suggestions
local (course)	12,453	2,060	685	human
RAF-DB	8,049	1,396	464	human
AffectNet	2,538	441	153	human
CK+	756	121	45	human
total	**36,327**	**6,310**	**3,688**	

7.2. Topic 2: Models, training and transfer

7.2.1. Theoretical basis

A convolutional network learns local filters whose responses are combined layer by layer into features such as eyebrow shape or mouth opening. Batch normalization [33] stabilizes training and dropout [34] reduces over-fitting. Residual connections [24] let gradients pass through deep networks; inverted residual blocks with depthwise separable convolutions [25] cut computation. **Transfer learning** re-uses features learned on a large source task: here ImageNet [23] gives general visual features, CK+ teaches clean, posed expressions of the six basic emotions, and the target data adapts the model to TV faces and adds the neutral class. Only the classifier is re-initialized at each step (6 → 7 outputs); every other tensor whose name and shape match is copied. Training in two stages, first the new head with a frozen backbone and then the upper blocks with smaller learning rates, protects the pretrained features from large early gradients.

7.2.2. Calculation/analysis

Compute. On the laptop RTX 3050 one epoch of the course dataset took about 45, 65 and 75 seconds for EmotionCNN, ResNet-18 and MobileNetV2, about 4 hours for all 9 midterm runs. The CK+ cross-validation (3 networks × 3 seeds × 10 folds = 90 runs) ran on Colab. Each mix run trained all 23 epochs in about 29 minutes (2026-11-09, 02:11–03:38 for three seeds).

Tuning on validation. The first ResNet-18 configuration (dropout 0.3, weight decay 1e-4) over-fitted: training macro-F1 0.95 against 0.69 on validation. AdamW scales its decoupled weight decay by the learning rate, so 1e-4 barely regularized; weight decay 5e-2 and dropout 0.5 kept validation macro-F1 (0.693) with a better-regularized model. For MobileNetV2, the final configuration (more unfrozen blocks, dropout 0.5, weight decay 5e-2) raised validation macro-F1 from 0.665 to 0.679. On CK+, which gives about 35 times fewer steps per epoch, MobileNetV2 needed five times larger learning rates (validation macro-F1 0.872 → 0.905).

Statistics. With 1,351 test images and about 70% accuracy, the 95% interval of the accuracy is about ±2.4 points, wider than the gap between the networks. Every result is therefore the mean ± standard deviation over three seeds, with a bootstrap interval from 1,000 resamples of the test set [35].

7.2.3. Results

Stage A, course dataset (midterm). The three networks are statistically tied (Table 14); averaging the nine runs with horizontal-flip test-time augmentation reaches 71.87%. EmotionCNN trained from scratch is as accurate as the pretrained networks on this large, 48×48 dataset.

Table 14: Stage A: course dataset test set (1,351 images, human labels), mean ± std over 3 seeds

Network	Accuracy (%)	Macro-F1	Macro-F1 95% CI
EmotionCNN	68.02 ± 1.04	0.679 ± 0.012	[0.654, 0.702]
ResNet-18	**68.32 ± 0.97**	**0.683 ± 0.010**	[0.658, 0.706]
MobileNetV2	67.83 ± 0.88	0.678 ± 0.009	[0.653, 0.701]
Ensemble of 9 runs + flip TTA	71.87	0.717	[0.694, 0.741]

Stage B, CK+ 10-fold cross-validation. Every image is tested once by a model that never saw its person (Table 15). ResNet-18 is the most accurate and the most stable; its lowest per-class F1 scores are sad (0.87), angry (0.88) and fear (0.90) (Figure 6).

Table 15: Stage B: CK+ subject-independent 10-fold CV (927 images, 6 classes), mean ± std over 3 seeds

Network	Pooled accuracy (%)	Pooled macro-F1	Worst / best fold (%)
EmotionCNN	90.36 ± 1.18	0.867 ± 0.015	76.7 / 100.0
ResNet-18	**94.28 ± 0.39**	**0.926 ± 0.006**	83.9 / 100.0
MobileNetV2	93.38 ± 0.53	0.903 ± 0.012	83.9 / 100.0

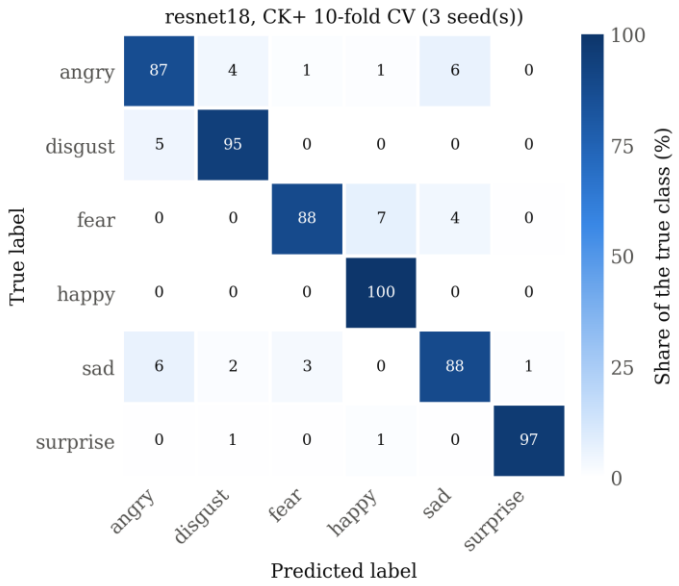


Figure 6: CK+ cross-validation confusion matrix of ResNet-18 (all seeds pooled, row-normalized)

Stages C and D, CK+ → VTV. Fine-tuned on the suggested TV labels, ResNet-18 again was best and most stable (Table 16). These numbers are agreement with the label-suggestion ensemble. The weakest classes were disgust (57%, confused with angry), fear (66%) and surprise (74%, confused with fear), the classes with the fewest real faces and the least reliable suggestions.

Table 16: Stages C and D: VTV test agreement with the suggested labels, mean ± std over 3 seeds

Network	C: agreement (%)	C: macro-F1	D: agreement (%)	D: macro-F1
EmotionCNN	80.29 ± 1.73	0.731 ± 0.027	77.31 ± 3.22	0.614 ± 0.042
ResNet-18	**83.64 ± 0.66**	**0.765 ± 0.003**	83.64 ± 0.66	0.701 ± 0.011
MobileNetV2	80.21 ± 0.88	0.733 ± 0.013	81.41 ± 0.85	0.663 ± 0.025

Diagnosis on human labels. When the first demo (stage D model) was tried on films, the recognition looked weak. Scored on RAF-DB faces with human labels, prepared exactly like the TV crops, the model reached only 53.4% on the RAF-DB test faces, below the ensemble that produced its labels (68.8%), and recalled 11% of surprise faces. High agreement had hidden a weak model: it had learned the ensemble's habits from a few hundred rare-class faces.

Stage E, VTV + course mix. Trained on the mix dataset, ResNet-18 (three seeds, each starting from the CK+ checkpoint of the same seed) converged in 14–17 epochs (Figure 7). On the full mix test set (3,688 faces) it scored 77.11 ± 0.59% / macro-F1 0.728 ± 0.004; per seed, see Table 17. Chapter 8 compares it with the earlier stages on human labels.

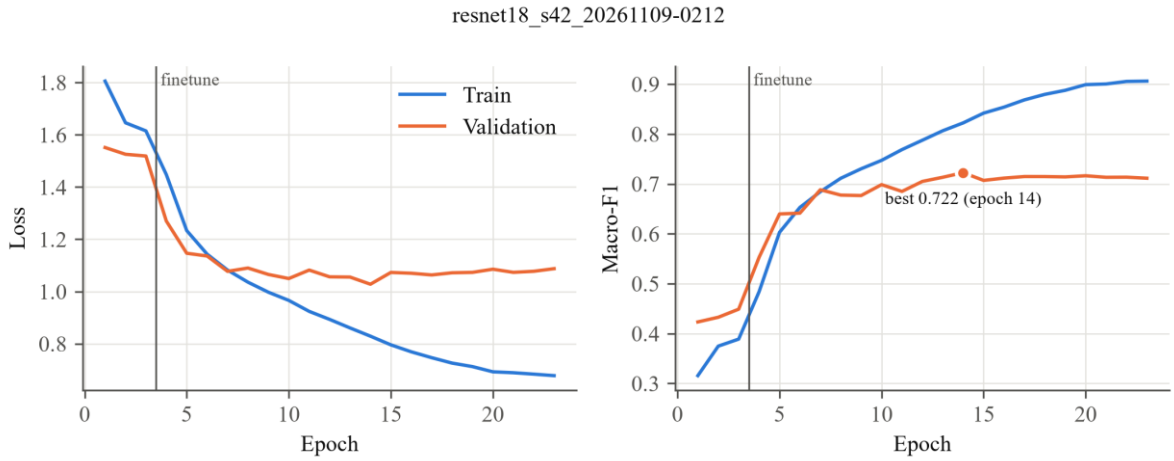


Figure 7: Learning curves of the stage E ResNet-18, seed 42 (loss and macro-F1)

Table 17: Stage E runs (ResNet-18); the application uses the seed with the best validation macro-F1

Seed	Best epoch	Val. macro-F1	Mix test acc. (%)	RAF-DB test acc. (%)	RAF-DB macro-F1
42	14	**0.722**	76.71	80.4	0.727
123	15	0.715	77.79	79.1	0.708
2024	17	0.711	76.84	78.7	0.705

7.3. Topic 3: Real-time application

7.3.1. Theoretical basis

A real-time pipeline is limited by its slowest stage. When stages run one after another, the frame rate is the inverse of their summed time; when two frames are in flight, the browser encodes frame n+1 while the server analyzes frame n, and the rate approaches the inverse of the slowest stage (Eq. (7)). Results can then arrive out of order, so the page drops any result older than the frame already shown and any result from before a seek. Seeking in large files works because the server answers HTTP range requests [42], so the browser loads only the part it plays.

$$FPS_{seq} = \frac{1000}{t_{enc} + t_{net} + t_{srv}}, \quad FPS_{pipe} \approx \frac{1000}{\max(t_{enc}, t_{srv})} \tag{7}$$

7.3.2. Calculation/analysis

On 1080p AV1 films, JPEG encoding of a full-resolution frame takes about 40 ms (t_{enc}) and the server about 46–49 ms (t_{srv}: decoding, detection, alignment, gates, and the network for every face). The first, sequential version reached 8 analyzed frames per second. With two frames in flight Eq. (7) gives an upper bound of about 20 frames per second, and 15–17 were measured (13 in scenes with many faces). Frames are sent at full resolution on purpose: the quality gates count pixels (eye distance ≥ 20 px), and a downscaled frame would turn many valid faces gray. The network itself is a small part of the budget: 2.6 ms per face on the GPU.

The whole-film scan reads one frame per step with a hidden second video element, without tracking. The application estimates 0.4 s per sampled frame; in the test for Figure 8 a 41-minute episode at one frame per 10 s (248 frames) took 70 s. The older OpenCV version of the demo, which decodes the video itself, runs at 17–38 frames per second depending on the number of faces.

7.3.3. Results

Figure 8 shows the application on a Vietnamese drama after a whole-film scan. The left column lists the files; the center shows the video with one confident face (angry 67%) and one gray face (turned); the timeline under it is colored by the scanned emotions and carries three user marks; the right panel shows the seven probabilities, the face crop next to the 48×48 crop the model saw, the user's marks and the strongest moments per emotion. Table 18 maps these functions to the two target fields. Figure 9 shows frames annotated by the demo on four Vietnamese programs.

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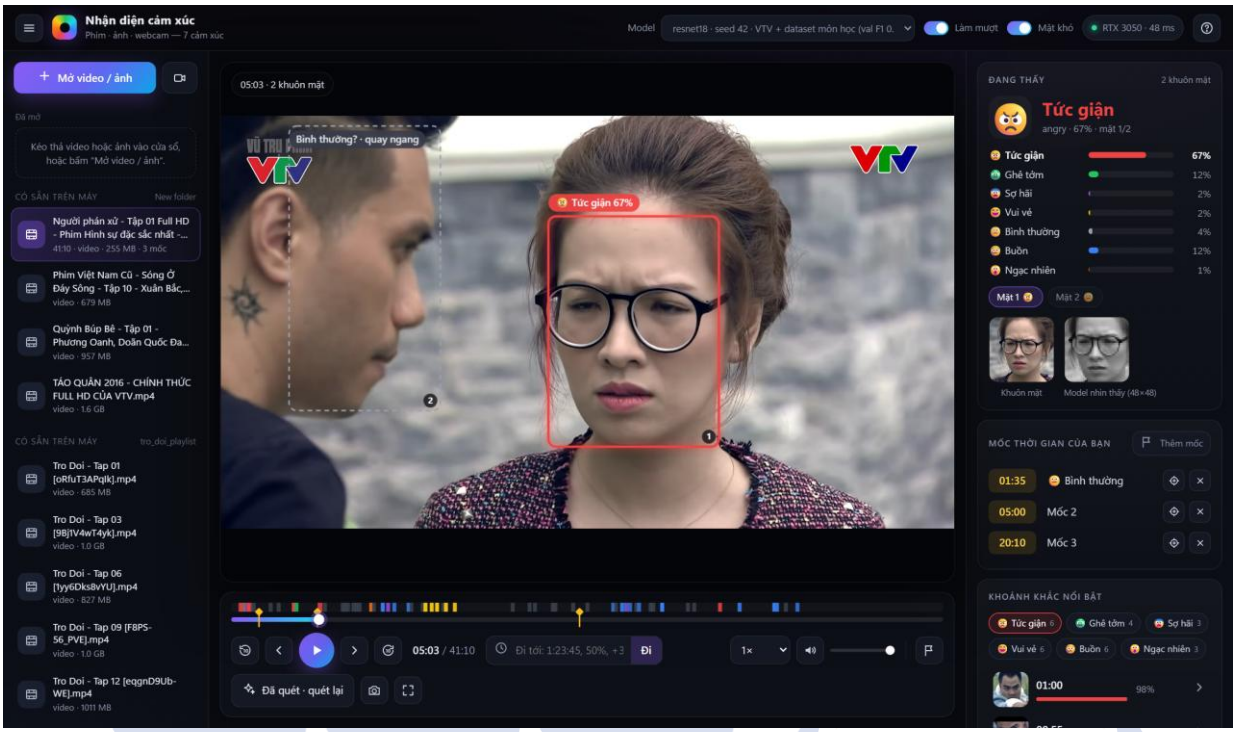


Figure 8: The browser application (interface in Vietnamese) after scanning a 41-minute episode of the drama Người Phán Xử

Table 18: Application functions and their use in medicine and art

Function	Medical use	Artistic use
Live view with probabilities	Follow a patient's expressions during a video consultation (with consent)	Feedback for an actor rehearsing in front of a webcam
Whole-file scan + emotion strip	Overview of affect across a recorded therapy session	Emotional arc of an episode; compare scenes or takes
Strongest moments per emotion	Jump to moments of distress for review	Find the strongest reactions for editing or teaching
Own time marks with labels	Clinician's notes at time points	Editor's or director's cue list
Gray faces and 48×48 crop	Shows when not to trust the output	Shows when lighting or angle defeats the model
Screenshot with boxes	Evidence for a case note	Material for a scene analysis

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Figure 9: Frames annotated by the demo on four Vietnamese programs (first stage E run); gray boxes mark faces outside the training conditions

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8. RESULTS AND DISCUSSION

8.1. Results

Table 19 collects the results of all stages. Only rows marked *human* are accuracy; rows marked *suggested* are agreement with the labels of the suggestion ensemble.

Table 19: Results of all stages (ResNet-18 unless noted; mean ± std over 3 seeds)

Stage	Test data	Labels	Accuracy / agreement (%)	Macro-F1
A Midterm	Course test, 1,351	human	68.32 ± 0.97 (ensemble 71.87)	0.683
B CK+ CV	CK+, 927 (10 folds)	human	94.28 ± 0.39	0.926
C CK+ → VTV	VTV test, 2,341	suggested	83.64 ± 0.66	0.765
D CK+ → VTV screened	RAF-DB test, 464	human	53.4 (seed 2024)	0.391
E CK+ → mix	RAF-DB test, 464	human	**79.4 ± 0.9**	**0.713**
E CK+ → mix	local test, 685	human	**60.8 ± 0.9**	**0.610**
E CK+ → mix	VTV test of C, 2,341	suggested	82.44 ± 1.24	0.748
Suggestion ensemble	RAF-DB test / local test	human	68.8 / 55.0	0.590 / 0.538

Figure 10 compares the models on the two human-labeled test sets that no model trained on, and Figure 11 shows the recall per class on RAF-DB. Adding about 4,200 human-labeled faces per rare class lifted surprise recall from 11% to 81%, disgust from 14% to 49%, angry from 30% to 65% and sad from 59% to 91%.

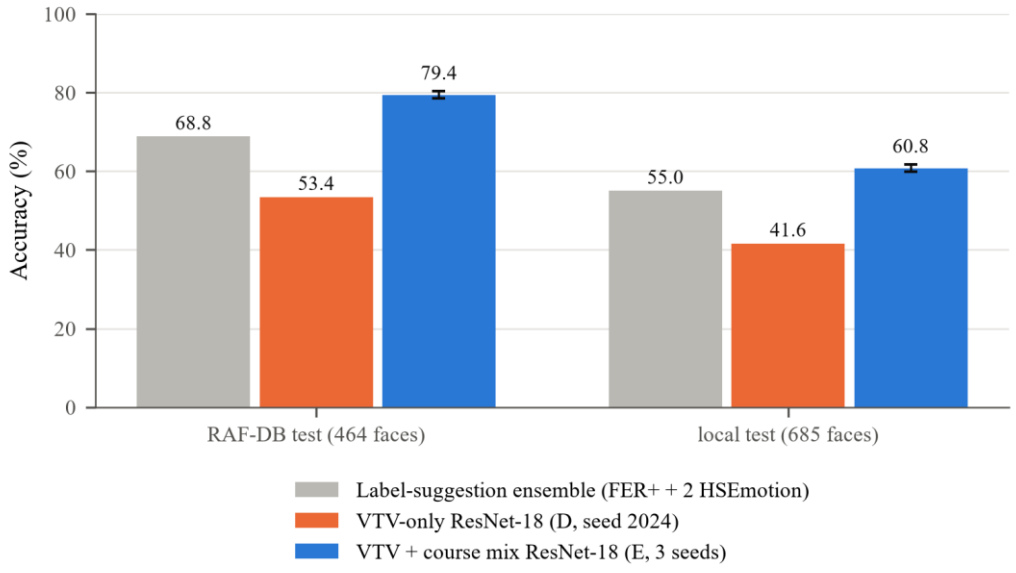


Figure 10: Accuracy on human-labeled test faces (error bars: std over 3 seeds)

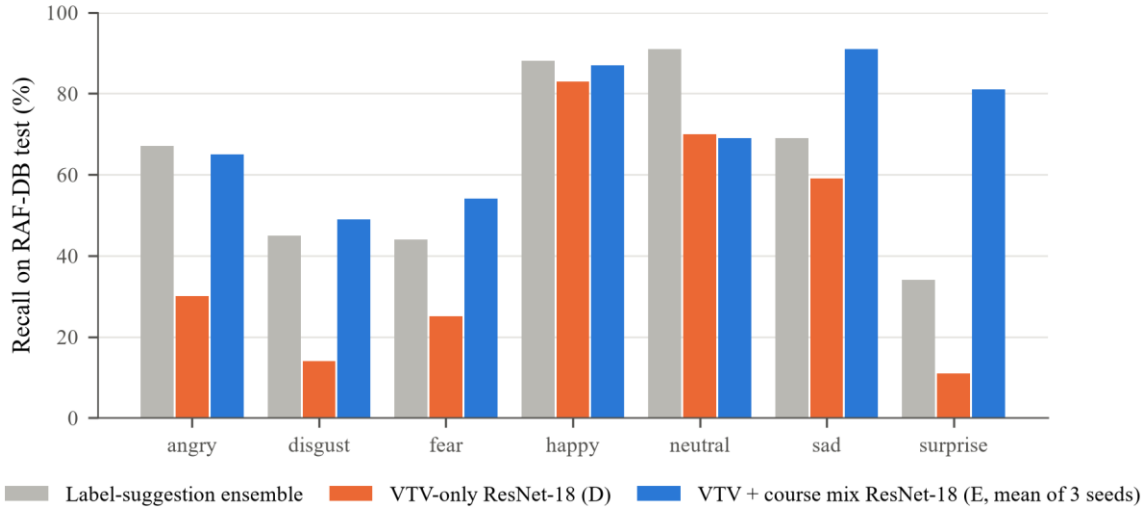


Figure 11: Recall per class on the RAF-DB test faces

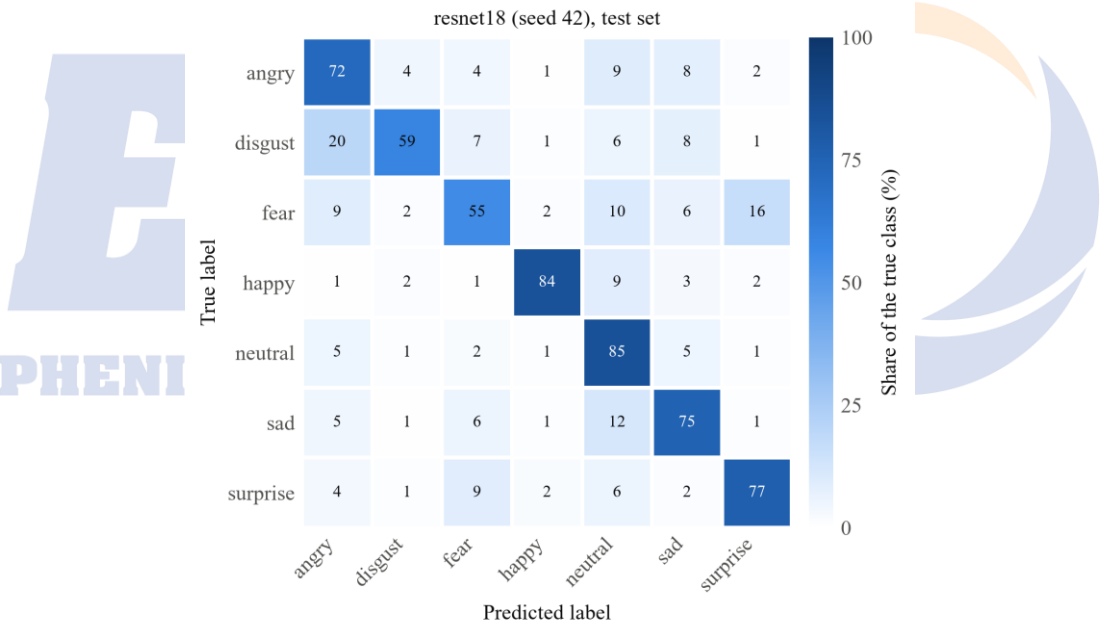


Figure 12: Confusion matrix of the application model (stage E, seed 42) on the full mix test set (3,688 faces; row-normalized)

Figure 12 shows the confusion matrix of the application model on the full mix test set. The largest confusions are disgust taken for angry (20%), fear for surprise (16%) and neutral (10%), and sad for neutral (12%): the same pairs on which human annotators disagree most.

8.2. Evaluation

Table 20: Objectives and expected results compared with what was achieved

Objective / expectation	Achieved	Status
CK+ subject-independent accuracy > 90%	94.3% (ResNet-18), 93.4%, 90.4%	met
≥ 70% on human-labeled in-the-wild faces, above the label suggestions	79.4% on RAF-DB test vs 68.8%; 60.8% on local test vs 55.0%	met on RAF-DB; local is harder
TV dataset > 10,000 faces, ≥ 1,000 per class, split by person	16,550 faces, 352 people; leak check 1.3% of test	met (labels not yet human-checked)
Three networks × three seeds with CIs	Stages A–C; stage E for ResNet-18 only	met / partly
≥ 10 analyzed frames/s at 1080p	13–17 frames/s (browser), 17–38 (OpenCV)	met
Seek, jump, marks, whole-film summary, local only	All in the browser application	met
Accuracy on Vietnamese TV faces	Not measurable yet (no human-checked TV test set)	open

The midterm accuracy of 68–72% on a mix of mostly in-the-wild 48×48 faces is in the range of human agreement on FER2013 (about 65% [19]). CK+ at 94% confirms that clear posed expressions are easy for all three networks. The 79.4% on RAF-DB test faces is the most relevant number for real use, with one caveat: model E trained on other RAF-DB and local faces, so part of the gain over model D is in-distribution. On the TV test set, model E agrees slightly less with the suggested labels than model C (82.4% vs 83.6%; the seed ranges touch), because it learned from human labels that differ from the ensemble's habits.

8.3. Discussion

Labels matter more than architecture. The three networks were within one point of each other on every dataset, but the choice of training labels moved RAF-DB accuracy by 26 points (53% → 79%). High agreement with model-suggested labels (83.6%) did not mean high accuracy: only a test set with human labels revealed the problem. For the same reason, removing doubtful faces (stage D) hurt, because it emptied the rare classes instead of fixing their labels.

Medicine. The application already supports reviewing a recorded session: the scan gives an overview, the moment list leads to the strongest expressions, and marks hold the clinician's notes. Before any clinical use, three things are missing: data from patients (with consent and ethics approval), classes that clinicians need (pain, flat affect, intensity), and a validation study against clinicians' ratings. The gray flags will be frequent in clinical video, where patients lie down or turn away.

Art. For film and theatre the tool is closer to use: the emotional arc of a whole episode is visible in about a minute, and editors or acting teachers can jump between the strongest reactions and compare

takes. Its labels describe what the face shows, which is exactly what an actor performs; the limits are crowd scenes, profiles and stylized make-up (see the comedy-show frames in Figure 9). The four demo programs were also sources of training faces, so a film outside the dataset would give a fairer impression.

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9. RISK ANALYSIS

9.1. Risk factor inputs

9.1.1. Risk factors

Table 21: Risk factors

ID	Risk	Cause	Consequence
R1	Noisy TV labels	Labels suggested by models, not checked by people	Model learns the suggesters' errors; TV accuracy unknown
R2	Domain shift in use	Patients, children, elderly people, hospital lighting, profiles	Lower accuracy than measured
R3	Over-interpretation or misuse	A label read as a feeling or a diagnosis; surveillance use	Wrong decisions about people; harm to trust
R4	Privacy breach	Patient or actor video copied or uploaded	Legal and ethical violation
R5	Bias	Unequal accuracy across age, gender, skin tone [41]	Unfair results for some groups
R6	Low memory on the laptop	16 GB RAM shared by data jobs and training	Killed runs (happened once)
R7	Cloud quota	Free Colab GPU time runs out	Delays (happened once; runs moved to the laptop)
R8	Licenses of data	TV episodes and research datasets	Material cannot be published
R9	Slow analysis	Many faces, 4K video, CPU-only computers	Missed short expressions
R10	Identity leakage	Same person in training and test	Inflated results

9.1.2. Probability and impact scales

Table 22: Probability and impact scales

Score	Probability P	Impact I
1	Rare: not expected during the project or first use	Negligible: no visible effect
2	Unlikely: could happen once	Minor: small delay or loss of accuracy
3	Possible: may happen a few times	Moderate: results or schedule clearly affected
4	Likely: expected to happen	Major: main results or users seriously affected
5	Almost certain: already happening	Severe: harm to people or the project fails

9.2. Risk analysis method

9.2.1. Scenarios method

For the highest risks, three scenarios were considered: the expected case, a worst case and the case after mitigation.

- **R1, noisy labels.** Expected: the model copies the ensemble's habits on TV faces (seen in stage D). Worst: a confident wrong label leads a user to a false conclusion. Mitigated: training mixes

in human-labeled faces (stage E) and all TV results are called agreement; a human-checked TV test set will give the real accuracy.

- **R2, domain shift.** Expected: accuracy in a clinic is below the 79% measured on RAF-DB. Worst: most faces are gray or wrong. Mitigated: quality gates flag faces outside the training conditions; a pilot with consented local data must come before any real use.
- **R3, misuse.** Expected: users read the top label as the person's feeling. Worst: the tool is used to judge employees or students. Mitigated: probabilities and crops are always shown, the guide states the limits, and such uses are excluded (they are prohibited under the EU AI Act [40]).

9.2.2. Risk score matrix

Each risk is scored by Eq. (8) and placed in the matrix of Figure 13: scores 1–4 are low, 5–9 medium, 10–16 high and 20–25 critical.

$$S = P \times I \tag{8}$$

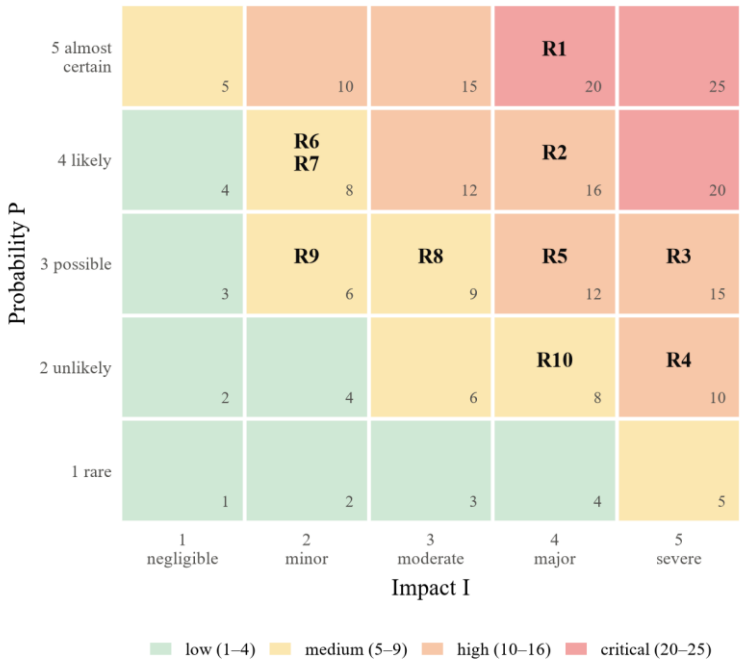


Figure 13: Risk score matrix (probability × impact)

9.3. Risk analysis result

9.3.1. Levels of risks

Table 23: Risk scores and levels

ID	Risk	P	I	S = P × I	Level
R1	Noisy TV labels	5	4	20	critical
R2	Domain shift in use	4	4	16	high
R3	Over-interpretation or misuse	3	5	15	high
R5	Bias	3	4	12	high
R4	Privacy breach	2	5	10	high
R8	Licenses of data	3	3	9	medium
R6	Low memory on the laptop	4	2	8	medium
R7	Cloud quota	4	2	8	medium
R10	Identity leakage	2	4	8	medium
R9	Slow analysis	3	2	6	medium

9.3.2. List of priority risks

Table 24: Priority risks and their mitigation

ID	Mitigation	Status
R1	Human review of the TV faces (review page logs every checked face); human-checked TV test subset; train with human-labeled faces (stage E)	partly done
R2	Quality gates with gray flags; pilot on consented local data before use; report accuracy per condition	gates done; pilot planned
R3	Probabilities and model crop always shown; guide and report state limits; no workplace or school use	done
R5	Measure accuracy by age, gender and skin tone on a labeled subset; add data where it is low	planned
R4	Local server bound to 127.0.0.1; no upload; consent for any recording; data kept private	done
R10	Splits by person with SFace clustering; leak check per build; de-duplication of the course data	done

10. FINANCIAL AND ECONOMIC REPORT

Table 25 lists the real costs of the project. Values marked *estimate* are calculated from stated assumptions, not from bills.

Table 25: Project costs

Item	Quantity / basis	Cost (VND)
Laptop (RTX 3050, 16 GB)	Already owned by a team member	0 (no purchase)
Google Colab	Free tier (T4 GPU)	0
Software	Open source: Python, PyTorch, OpenCV, Flask, scikit-learn	0
Datasets and models	Public research datasets and models; TV episodes publicly available	0
Electricity (estimate)	≈ 20 h of GPU load × ≈ 0.13 kW ≈ 2.6 kWh at ≈ 3,000 VND/kWh	≈ 8,000
Total cash cost		**≈ 8,000 (under 1 USD)**

The 20 GPU hours are estimated from the logged laptop jobs: about 4 h for the midterm runs, about 9 h of face extraction from the TV episodes, about 3.5 h for the laptop transfer and mix runs, and about 3 h of tests and the application. The students' time (about seven weeks of part-time work) is the real cost of the project and is not priced here.

Deployment. A clinic, school or studio can run the application on an existing Windows PC. Without a GPU the network needs about 17 ms per face on the CPU, so the application stays usable at a lower frame rate; a desktop with an entry-level NVIDIA GPU gives the speed reported here. The software has no license fee, in contrast with the commercial tools of Table 2. The main cost of a real deployment would be the validation study and the human labeling of local data, not hardware.

11. CONCLUSIONS

This project built a complete facial emotion recognition system for Vietnamese faces, from data to a real-time application, aimed at reviewing recordings in medicine and art.

- Three networks were compared under one pipeline with three seeds each. ResNet-18 was the most accurate and most stable: 68.3% on the course test set, 94.3% in subject-independent CK+ cross-validation.
- A dataset of 16,550 Vietnamese TV faces of 352 people was built with a reproducible pipeline that aligns faces like CK+, filters them with quality gates and splits them by person.
- Measuring on human labels exposed a model that only agreed with its label suggesters (53% on RAF-DB). Training on TV faces together with human-labeled course faces raised this to 79.4 ± 0.9%, with the largest gains on the rare emotions.
- A local browser application opens any video or image, seeks and jumps to any time, keeps marks, scans a whole film for its strongest moments and runs at 13–17 analyzed frames per second on 1080p video.

The main open point is accuracy on Vietnamese TV faces, which needs a human-checked test set. Future work, in order: (1) human review of the TV labels and retraining; (2) Grad-CAM [29] maps for the three networks to check that the models look at eyes, brows and mouth; (3) a faster MobileNetV2 version of the mix model; (4) more real faces for the rare classes; (5) valence, arousal and intensity outputs; and (6) pilot studies with clinicians and film professionals, with consent and ethics approval.

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12. STANDARDS

Table 26 lists the standards and regulations that apply to the project or to a later deployment.

Table 26: Applicable standards and regulations

Standard / regulation	Topic	Relevance
IEEE Std 7000-2021 [43]	Ethical concerns in system design	Method used to list stakeholder values (privacy, honesty, fairness) in Section 5.7
IEEE Std 7014-2024 [44]	Ethics of emulated empathy in AI systems	Directly about systems that recognize human emotion
ISO/IEC 23894:2023 [45]	AI risk management	Structure of the risk analysis in Chapter 9
ISO/IEC 42001:2023 [46]	AI management systems	For an organization that deploys the tool
IEC 62304 [47], ISO 14971 [48]	Medical device software and risk management	Required if the tool ever becomes a medical device; not claimed now
Decree 13/2023/ND-CP [8], Law 91/2025/QH15 [49]	Vietnamese personal data protection	Face images are personal data; consent and local processing
GDPR [50], EU AI Act [40]	EU data protection and AI rules	Biometric data; emotion recognition banned in workplaces and schools except medical or safety use
RFC 9110 [42]; JPEG, PNG, MP4	HTTP range requests; image and video formats	Seeking in large files; frame and screenshot formats
IEEE reference style	Citations and references	Format of the references in Chapter 13

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14. APPENDICES

Appendix A: Main commands (from the project folder `D:\Project\facial emotion`).

Table 27: Main commands

Task	Command
Open the application	`app_nhan_dien.bat` or `python -m src.demo.app`
OpenCV demo with a file dialog	`python -m src.demo.webcam --pick`
Train one run	`python -m src.train --config configs/transfer/resnet18_mix.yaml --seed 42 --init <CK+ run>`
Evaluate on the test set	`python -m src.evaluate --run <run directory>`
Score against human labels	`python -m src.eval_wild --runs "runs/transfer_mix/resnet18_s*" --splits test`
Collect results	`python -m src.aggregate --runs runs/transfer_mix --out reports/transfer_mix_results.csv`
TV dataset pipeline	`python -m src.data.vtv_pipeline --status`, then `--detach`

Appendix B: Keyboard shortcuts of the application.

Table 28: Keyboard shortcuts of the browser application

Key	Action	Key	Action
Space / K	play, pause	B	add a time mark
← / →	back / forward 5 s	G or /	type a time to jump to
Shift + ← / →	back / forward 1 min	S	screenshot with boxes
J / L	back / forward 10 s	F	full screen
, / .	one frame back / forward	M	mute
0 … 9	jump to 0 … 90%	PgUp / PgDn	previous / next file
O	open a file	?	help

Appendix C: Deliverables. The project folder contains the code (`src/`), the configurations (`configs/`), the reports and figures (`reports/`), and a summary folder `tong_hop_ket_qua/` with copies of every result table, the training runs of all stages, the application with its guide, and the application model (`models/resnet18_mix_s42_app_model/`). The TV frames and the research datasets are covered by licenses and are kept private.